



Face Recognition in Group Photos

This project detects and recognizes known people in a group image using Python, OpenCV, and the `face_recognition` library.

It draws bounding boxes around detected faces and labels each recognized person by name.



Features

- Detect multiple faces in a group photo
 - Recognize known individuals
 - Label faces with names
 - Save the final annotated image
 - Clean and modular project structure
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How It Works

1. Load known face images from `data/known_faces/`
 2. Extract face encodings for each person
 3. Load the target group image
 4. Detect all faces in the image
 5. Compare each detected face with known encodings
 6. Draw bounding boxes and labels
 7. Save the final result to `output/result.jpg`
-



Project Structure

```
face-recognition-project/  
|  
├─ data/
```

```
|   ├── known_faces/
|   |   ├── mohsen.jpg
|   |   ├── komeil.jpg
|   |   └── javad.jpg
|   └── group.jpg
|
|── demo/
|   └── result.jpg
|
|── output/
|
|── main.py
|── requirements.txt
|── README.md
```

🔧 Installation

Clone the repository:

```
git clone https://github.com/your-username/face-recognition-project.git
cd face-recognition-project
```

Install dependencies:

```
pip install -r requirements.txt
```

🎬 Usage

Run the script:

```
python main.py
```

The output image will be saved in:

```
output/result.jpg
```

A window will also open showing the result.

Demo

Input: Group photo

Output: Faces detected and labeled

You can see an example result inside the `demo/` folder.

Requirements

- Python 3.8+
 - OpenCV
 - face-recognition
 - dlib
 - numpy
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Notes

- Each image inside `data/known_faces/` should contain only one clear face.
 - File names are mapped to display names inside the `PERSONS_MAP` dictionary.
 - Images without detectable faces are skipped automatically.
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License

This project is for educational and portfolio purposes.